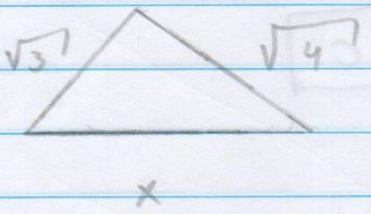
Beatriz Bastos Borges Nº 03 CT11350

Tarefa Básica

1.

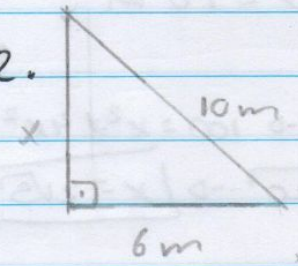


$x^2 = (\sqrt{3})^2 + (\sqrt{4})^2 \rightarrow x^2 = 7$

$\boxed{x = \sqrt{7}}$

$\boxed{\text{Alternativa B}}$

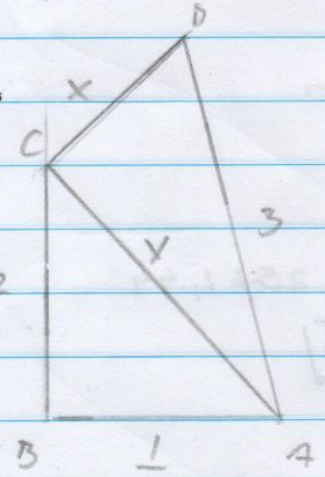
2.



$10^2 = 6^2 + x^2 \rightarrow x^2 = 100 - 36$

$x = \sqrt{64} \rightarrow \boxed{x = 8m}$

3.



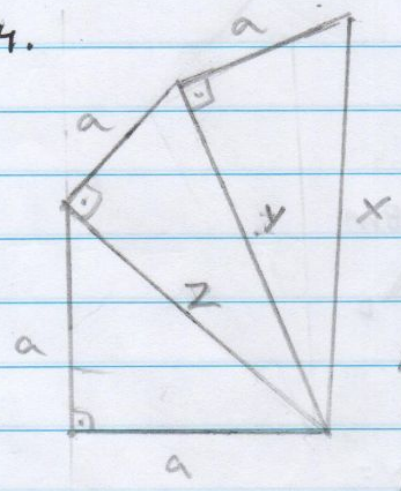
$y^2 = 2^2 + 1^2 \rightarrow y = \sqrt{5}$

$3^2 = (\sqrt{5})^2 + x^2 \rightarrow x^2 = 9 - 5$

$x = \sqrt{4} \rightarrow \boxed{x = 2}$

$\boxed{\text{Alternativa 2}}$

4.



$z^2 = a^2 + a^2 \rightarrow z = \sqrt{2a^2}$

$y^2 = a^2 + (\sqrt{2a^2})^2 \rightarrow y = \sqrt{3a^2}$

$x^2 = a^2 + (3a^2) \rightarrow x = \sqrt{4a^2} \rightarrow \boxed{x = 2a}$

$\boxed{\text{Alternativa B}}$

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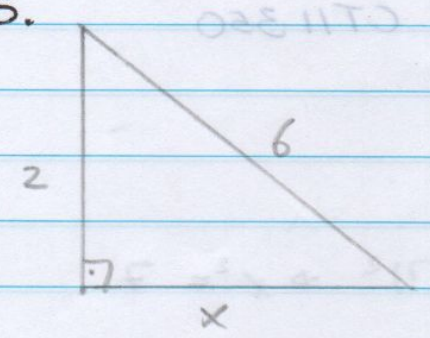
5.

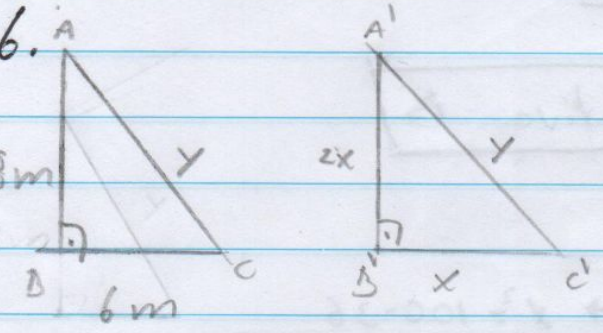


$6^2 = 2^2 + x^2 \rightarrow x^2 = 36 - 4$

$x = \sqrt{32} \rightarrow \boxed{x = 4\sqrt{2}}$

$\boxed{\text{Alternativa C}}$

6.



$y^2 = 8^2 + 6^2 \rightarrow y^2 = 64 + 36$

$y = \sqrt{100} \rightarrow y = 10\ m$

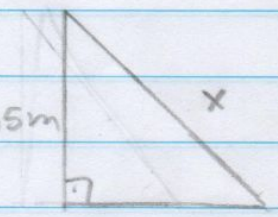
$10^2 = x^2 + (2x)^2 \rightarrow 100 = x^2 + 4x^2$

$\frac{100}{5} = x^2 \rightarrow x = \sqrt{20} \rightarrow \boxed{x = 2\sqrt{5}}$

$\boxed{\text{Alternativa A}}$

7. aranha → $16 \cdot 5 = 80$ cm ou $0,8$ m

formiga → $10 \cdot 5 = 50$ cm ou $0,5$ m

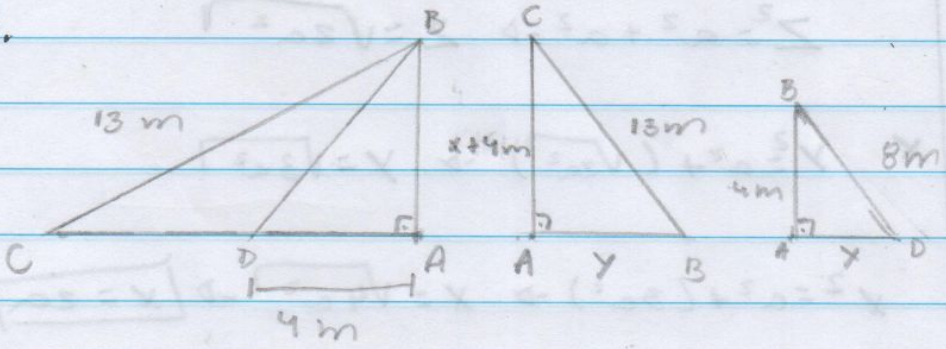


$x^2 = 0,5^2 + 1,2^2 \rightarrow x^2 = 0,25 + 1,44$

$x = \sqrt{1,69} \rightarrow \boxed{x = 1,3\ m}$

$\boxed{\text{Alternativa B}}$

8.



$8^2 = 4^2 + y^2 \rightarrow y^2 = 64 - 16 \rightarrow y = \sqrt{48}$

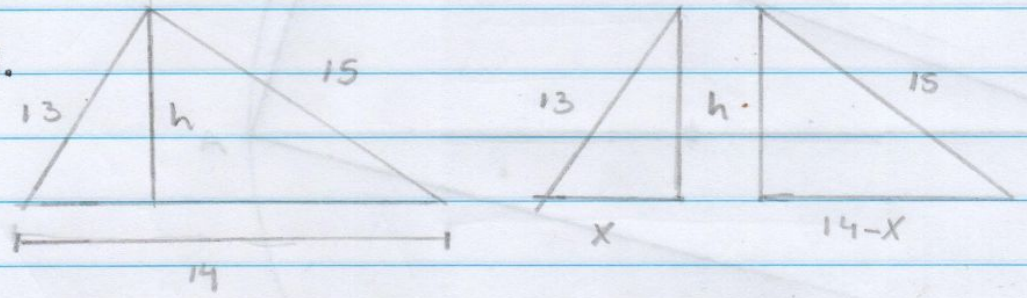
$13^2 = (\sqrt{48})^2 + (4+x)^2 \rightarrow 0 = 48 + 16 + 8x + x^2 - 169 \rightarrow 0 = x^2 + 8x - 105$

S $7 + (-15) = (-8)$ $\boxed{x' = 7m}$

P $7 \cdot (-15) = (-105)$ $x'' = (-105) \rightarrow$ não convém

$\boxed{\text{Alternativa D}}$

9.



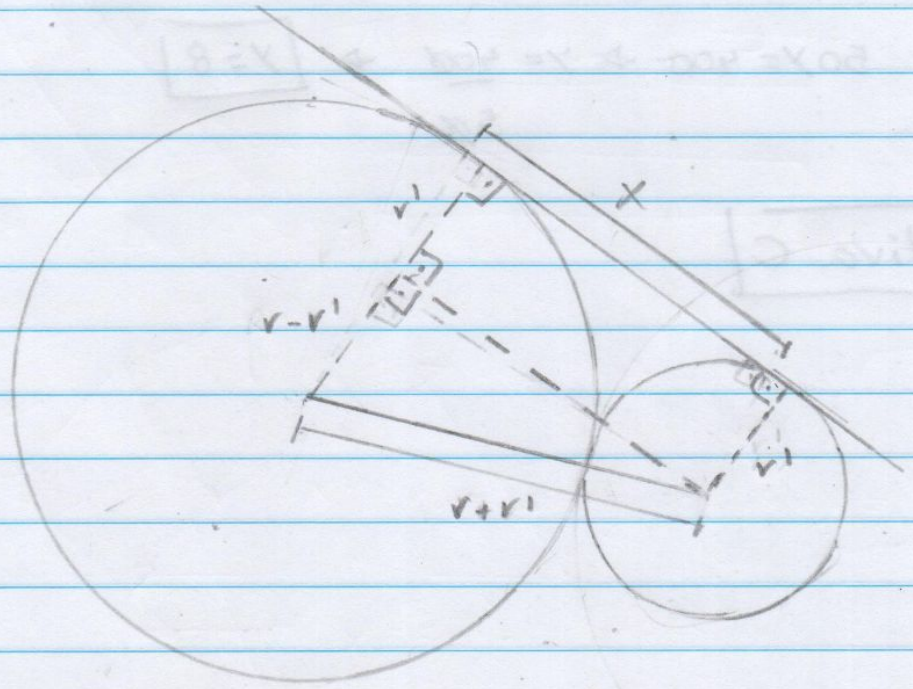
$15^2 = h^2 + (14-x)^2 \rightarrow 225 = h^2 + 196 - 28x + x^2$

$h^2 = 225 - 196 + 28x - x^2 \rightarrow h^2 = 29 + 28x - x^2$

$13^2 = h^2 + x^2 \rightarrow 169 = 29 + 28x - x^2 + x^2 \rightarrow 140 = 28x \rightarrow x = 5$

$13^2 = h^2 + 5^2 \rightarrow h^2 = 169 - 25 \rightarrow h = \sqrt{144} \rightarrow \boxed{h = 12}$

10.



$(r+r')^2 = (r-r')^2 + x^2 \rightarrow r^2 + 2rr' + r'^2 = r^2 - 2rr' + r'^2 + x^2$

$x^2 = 4rr' \rightarrow x = \sqrt{4rr'} \rightarrow \boxed{x = 2\sqrt{rr'}}$

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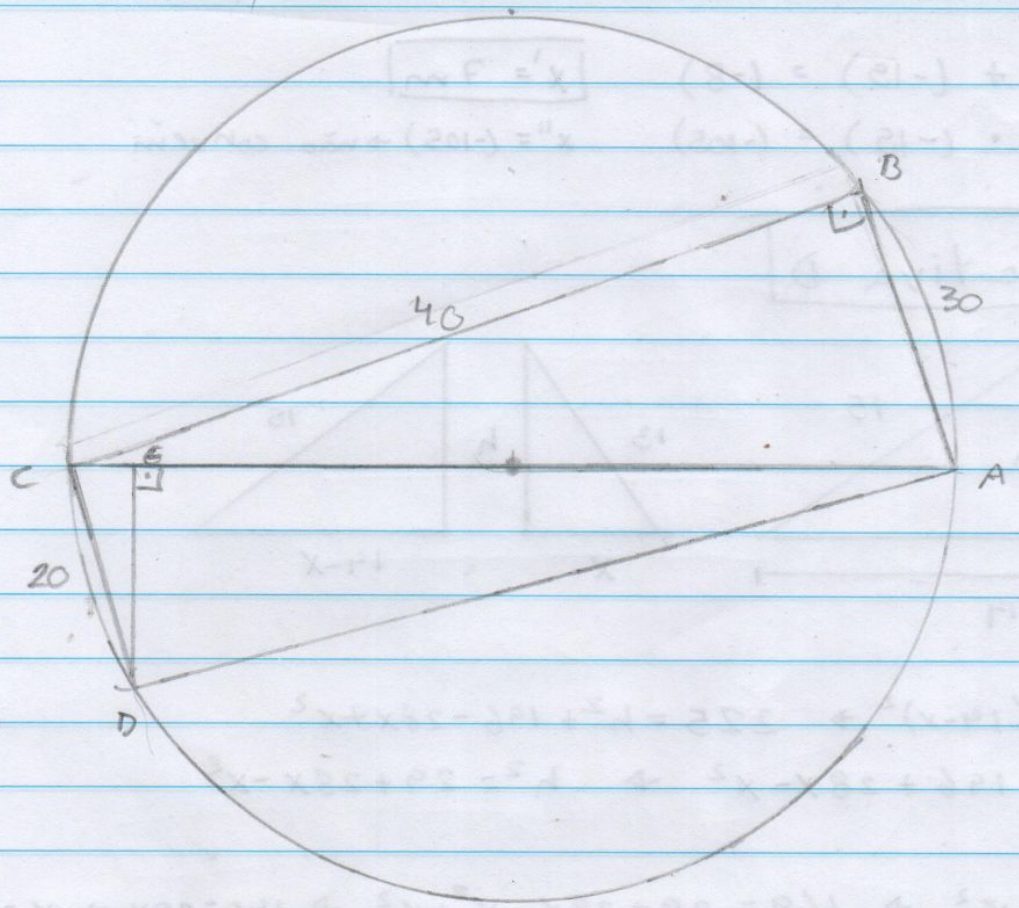
11.



$x^2 = 40^2 + 30^2 \rightarrow x^2 = 1600 + 900 \rightarrow x = \sqrt{2500} \rightarrow x = 50$

$\frac{20}{y} = \frac{50}{20} = 50y = 400 \rightarrow y = \frac{40\cancel{0}}{5\cancel{0}} \rightarrow \boxed{y = 8}$

$\boxed{\text{Alternativa C}}$